

Project Overview

This project focuses on using historical workforce data builds upon the 'VOC-opvarenden' database, aiming to examine is there relationship between VOC's workforce structure and the Company's eventual decline. The VOC, a powerful monopoly granted by the States General to trade in Asia from 1602 to 1800, has been studied extensively, with previous research citing factors like warfare with England, financial debt, and problematic workforce patterns as contributors to its collapse. Given that our dataset consists of detailed VOC worker records covering the period from 1700 to 1780, we have chosen to concentrate specifically on analyzing these workforce dynamics. Our core Research Question (RQ) is: Can the fall of the VOC be predicted or foreshadowed by patterns observed in this workforce data? Current scholarship already highlights several workforce-related issues that contributed to the Company's decay, including a growing reliance on foreign recruits, persistently high levels of mortality and desertion, and limited opportunities for career advancement that collectively created significant instability within the organization. Based on these findings, our central thesis statement argues that the VOC's decline can somehow be foreshadowed in its workforce patterns. We aim to rigorously analyze the dataset to validate this hypothesis while also striving to identify any new or under-examined workforce factors that may have played a critical role in the Company's final collapse.

Data Acquisition

The dataset consists of an enhanced version of the original 'VOC-opvarenden' database (VOC: Opvarenden, z.d.) curated by the International Institute of Social History (Petram et al., 2024). This collection is developed from 774,200 muster records and has been enriched through the disambiguation of individual records, the standardization of 44,152 unique placenames and the addition of wage details and rank structure. The VOC operated between 1602 and 1800 and during this time the company documented many reports, instructions and pay ledgers with salaries of seamen and the enlisted soldiers. Of the records around 25 million pages survived and from the ledgers of 3,248 outbound voyages (mostly from the eighteenth century) with 774,200 muster records did so as well. These records were digitized and formed into a database titled 'VOC-opvarenden'. The work of Petram et al. (2024) consists of enhancing that database to optimize its utility for future research. For this optimization process the authors mention different challenges of the database. First being its focus on employments instead of individuals which splintered the data for those who reentered in the VOC. Secondly, the database is limited for geographical analysis since crew member's places of origin were phonetically noted which resulted in a wide range of spelling for the same place names. To solve this issue, Petram and colleagues standardized these place names. To further enrich the database the authors linked individuals to their respective records, organized ranks in the records and integrated wage data to the dataset.

For this project, the data were filtered to contracts that started between 1700 and 1780, resulting in 676,934 records. This time frame was chosen because it represents the most complete and consistent part of the dataset. Focusing on this period allowed us to avoid gaps and irregularities from the earlier and later decades, ensuring more reliable results for our research question.

Data Acquisition: Data Preprocessing Steps

The preprocessing was carried out using Python (version 3.11) with the libraries pandas, numpy, and matplotlib. The main steps were as follows:

1. Filtering and temporal scope
The contracts were filtered to include only those starting between 1700 and 1780. A new variable, decade, was created to facilitate trend analysis over time.
2. Region mapping
Each contract's place of origin was matched with the standardized toponym table (voc_places_standardized.csv) that contained region codes (A–I). When a record lacked a standardized match but included a recognizable Dutch city name, such as Amsterdam or Zeeland, it was manually assigned to the Dutch Republic region. After this step, the dataset achieved 81.6 percent region coverage, while 99.5 percent of all records still contained valid origin information.
3. Rank classification and seniority estimation
The detailed ranks listed in voc_ranks.csv were grouped into six main parent categories: Sea, Ship, Trade, Medical, Military, and Other. Wage information was used to estimate relative seniority levels. These were divided into five quintiles and later collapsed into a binary variable, high rank, distinguishing higher from lower-ranking positions.
4. Outcome recoding
The variable describing how each contract ended was inconsistent across the raw dataset. To make analysis possible, the outcomes were reclassified into four groups: Death, Repatriated, Attrition, and Unknown.
5. Recruitment identification
For each unique individual (identified by person_cluster_id), the earliest recorded contract was marked as a first contract. This allowed us to track recruitment trends over time and by rank.
6. Diagnostics and data integrity
During processing, coverage statistics were printed and reviewed. Summary tables were exported to confirm that region, rank, and outcome categories aligned with expectations and historical plausibility.

The final processed dataset was saved as contracts_clean.csv. Additional summary tables, such as rank_counts.csv, region_counts.csv, and outcome_counts.csv, were used to verify distributional accuracy before analysis.

Methodology

The methodological design combines descriptive statistical analysis with exploratory predictive modeling to investigate how workforce characteristics within the VOC evolved between 1700 and 1780. The purpose was not to forecast the company's decline but to identify quantifiable patterns that may point to structural imbalances in its labor system.

Two analytical layers were implemented:

1. Descriptive analysis was used to visualize distributions and relationships between region, rank, outcome, and time. This step was essential for understanding overall patterns and trends before applying any predictive model.

2. Predictive modeling was used to examine whether certain workforce attributes could help predict contract outcomes, such as death or repatriation.

The decision to combine these two approaches was made to integrate interpretability with analytical rigor. Descriptive results provided historical clarity, while modeling introduced an additional quantitative perspective that could test whether observed patterns were statistically consistent.

We first selected the Random Forest classifier because it performs well with large, imbalanced datasets and can capture non-linear relationships between features—conditions that fit the VOC data. It was expected to reveal complex interactions, such as how rank and region jointly influenced survival.

After running initial tests, we added Logistic Regression as a comparison model to provide a simpler, statistically interpretable benchmark. Although Logistic Regression assumes linear relationships, it offers clear coefficient-based interpretations that can be useful for understanding direction and strength of associations.

Comparing both approaches allowed us to balance predictive performance with interpretive value. Random Forest was suited to uncovering subtle structural patterns, while Logistic Regression provided a transparent baseline for verifying those findings.

Alternative techniques, such as unsupervised clustering or time-series models, were considered but rejected because they would have obscured the historical meaning of the variables or required information that was not consistently available across the dataset.

Team collaboration within the methodology

Feruza was responsible for the full data workflow, including preprocessing, descriptive analysis, visualization, and modeling. Dunya and Emily worked on the historical literature, sustainability context, and narrative interpretation. Together, we combined technical findings with historical understanding, ensuring that the quantitative results were grounded in relevant context.

Workflow Steps

The project workflow followed a clear, reproducible structure, organized into three Python scripts:

1. Data preparation (01_cleaning.py)
 - Filtered contracts to 1700–1780.
 - Mapped places of origin to standardized regions.
 - Created rank categories and derived seniority variables.
 - Simplified outcomes and marked first contracts.
 - Exported processed data and diagnostic summaries.
2. Descriptive analysis (02_descriptives.py)
 - Generated cross-tabulations by region, rank, and outcome.

- Produced trend tables such as `trend_by_decade.csv` to visualize death, attrition, and recruitment changes over time.
 - Created visual outputs stored in the `/figures` folder, including:
 - `fig_region_by_rank_parent.png`
 - `fig_death_vs_recruitment_by_decade.png`
 - `fig_death_vs_recruitment_by_rank.png`
 - `fig_outcome_trends_plus_recruitment_by_rank.png`
3. Modeling (`03_models.py`)
- Encoded categorical variables and divided the data into training (507,700) and test (169,234) sets.
 - Trained Random Forest and Logistic Regression models sequentially. Random Forest was tested first for its strength in handling data imbalance, while Logistic Regression was added later as a comparative baseline.
 - Evaluated models using accuracy, precision, recall, and confusion matrices.
 - Exported key visualizations:
 - `fig_logit_confusion_matrix.png`
 - `fig_rf_feature_importance.png`
 - `fig_rf_confusion_matrix.png`
4. Interpretation and validation
- Compared the results of the two models.
 - Random Forest provided more balanced classification and more interpretable feature importance, even though its overall accuracy was slightly lower.
 - Rank and region consistently appeared as the most influential features across both models.

Challenges and Solutions

The first challenge we met was “navigating research design and focus”. Our preliminary ideas, which focused narrowly on death rate and rank, were deemed too similar to previous semester topics after discussion with the professor. This necessitated a complete pivot. The constraint of the dataset—which focused exclusively on employee data—made it difficult to “think out of the box,” leading us repeatedly back to internal queries such as “which group survives?” or “what work does each group perform?” The breakthrough came when we consciously shifted our perspective to view the data through the lens of the “company’s” overall health and structural strategy, ultimately leading to our current question focused on prediction and the workforce as a foreshadowing indicator of decline.

The second challenge we met was “reconciling disparate methodological mindsets”. Feruza, from an AI/computational background, prioritized quantitative accuracy, while Dunya and Emily, from a social science perspective, emphasized historical context. This difference of opinion was most pronounced during the discussion about the project’s time range. While including the full dataset timeline was tempting for context, we found the data completeness was lacking during the VOC’s true beginning and end phases. Given our RQ’s objective to “predict” the fall—a goal dependent on high accuracy—we ultimately agreed to a compromise: focusing solely on the 1700–1780 range. This ensured data integrity and

completeness for modeling purposes, while acknowledging that the excluded periods remain valuable for future, context-focused research.

The third challenge was “selecting and refining analytical methodology”. Initially, aggregated data suggested the VOC was a healthy company from an HR perspective, as recruitment rates consistently exceeded death rates. This was counter-intuitive to our hypothesis. Following the professor's advice, we performed a deeper analysis, disaggregating the data by rank. This revealed that the overall recruitment success was highly misleading: the VOC only managed to recruit sufficient numbers in the "sea" rank, failing to match losses in other crucial ranks. This discovery provided a strong foundation for our central argument: the VOC suffered from a critical internal structural imbalance and a flawed hiring strategy.

Last but not least, group cohesion and cooperation were managed effectively. We maintained constant communication through WhatsApp and held focused meetings approximately every two weeks. By establishing clear individual goals and planning around our free time, we fostered a strong foundation of mutual respect and understanding. This proactive approach to team management was essential in ensuring an effective and efficient workflow.

Ethical Considerations

Modern day ethical considerations like for example informed consent are impossible to apply to a dataset like this with personal records tracing back over three hundred years from people who have ceased to exist. Privacy considerations are therefore impossible to incorporate in our research documentation. These data are collected only from VOC workers who signed formal contracts in the Netherlands. Another consideration to make is that the data have a strong Dutch bias. Taking into account that the VOC also employed many Chinese and other Asian sailors during the 1780s due to a severe shortage of sailors, but that these Asian seamen were rarely reported in the seventeenth century (Lucassen, 2004), (Blussé, 2015). Besides the incomplete records excluding foreign workers the data are also generated solely by Dutch researchers which reinforces the Dutch narrative eliminating other perspectives. Managing these issues is accompanied by making decisions. One of these decisions exists of accepting that working with historical data always generates incomplete results that are not a hundred percent legitimate. In dealing with privacy concerns a decision was made that the people in this dataset are long gone and finding descendants would be too costly and unsustainable so we decided to accept that we could not realize this concern. For contesting the Dutch bias, we were to some degree able to withstand this complication. Working in an interdisciplinary setting automatically encourages discussion and allows for different perspectives in a research project. In our research we worked with three different perspectives on the VOC as a company and the data in the database. The first perspective is a completely blank one. Coming from Uzbekistan Feruza Bakhtiyorova never heard of the VOC before and thus got to approach the project from a blank standpoint. Secondly there is another Dutch perspective. Coming from Enkhuizen (one of the VOC cities in The Netherlands) Dunya Boon received a lot of education on this time period including the glory years of the company while also noting the black page of this time in Dutch history. Thirdly, Emily Li provided different insights about the VOC in Taiwan. These insights consisted that in the Taiwan education Li received no slave trading done by the VOC was ever mentioned. Instead of highlighting the dark sides of the company the Li mentioned the VOC seafarers to

be considered smart and not cruel since they focused on trading with Taiwan instead of killing indigenous inhabitants. Approaching our research from a blank slate of data, the Dutch taught history of the company and the diverse insights from a Taiwan perspective resulted in an inclusive approach of the data and an overall comprehensive research project.

Results

Descriptive findings

The descriptive analysis revealed consistent patterns of imbalance across the VOC workforce between 1700 and 1780.

- Regional composition (Figure: Region by Rank)
Dutch workers dominated nearly all ranks except the military, where they represented about 34 percent. This lower share likely results from incomplete or inconsistent records rather than a real historical difference.
- Mortality and recruitment trends (Figure: Death Rate vs. Recruitment by Decade)
Across decades, death rates remained high while recruitment did not rise to replace losses. The imbalance between mortality and inflow indicates long-term workforce strain.
- Rank-specific recruitment (Figure: Death vs. Recruitment by Rank)
The company did not recruit more heavily in the ranks suffering the highest losses. This misalignment suggests weak organizational adaptation to ongoing attrition.
- Outcome trends by rank (Figure: Outcome Rates and Recruitment Share by Decade)
The small multiple plots show clear contrasts between ranks.
 - Medical ranks had more repatriations than deaths.
 - Military ranks had the highest death rates and the lowest recruitment.
 - Sea ranks maintained high recruitment and high death rates simultaneously.
 - Ship ranks shifted from higher repatriation to higher mortality later in the century.
 - Trade ranks consistently experienced death as the dominant outcome.Overall, death remained the most frequent outcome across most ranks, while recruitment stayed comparatively low.

These trends collectively suggest that the VOC's staffing system was unable to balance losses with sufficient new inflow.

Modeling findings

Although Logistic Regression achieved a slightly higher accuracy (0.44 versus 0.39 for Random Forest), its predictions were heavily weighted toward the dominant outcomes (Death and Repatriated), producing a less balanced view of the data.

The Random Forest model, while marginally less accurate overall, offered a richer and more historically meaningful interpretation of workforce patterns. It successfully captured variation across all outcome classes, including the smaller but significant category of Attrition, and its feature importance rankings mirrored realistic historical dynamics—such as the impact of rank, region, and contract decade.

Therefore, Random Forest was considered the more reliable model overall, as it reflected workforce imbalance more convincingly, aligned with historical evidence, and provided deeper interpretive insights beyond numerical accuracy.

Although neither model was designed to predict the company's collapse directly, the results confirmed that workforce outcomes were not random. They followed clear structural patterns that reflected broader inequalities and inefficiencies in how the VOC managed its personnel.

Interpretation

The combined evidence from descriptive and modeling analyses shows that the VOC's workforce structure was marked by persistent high mortality, low recruitment, and a mismatch between where losses occurred and where replacements were made. These internal imbalances weakened the sustainability of the organization's human resources.

While the project does not claim to predict the company's fall, the data strongly suggest that systemic workforce instability was a contributing factor to the VOC's long-term decline.

Documentation and Sustainability

All data processing and analysis in this project were designed to be transparent, reproducible, and easy to maintain. The complete workflow is hosted on **GitHub** under the repository

[Intro-to-DH-SA---Group-2---The-Dutch-East-India-Company-s-Eighteenth-Century-Workforce](#)

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The repository contains all relevant files and is organized into clearly structured folders:

- `data_raw/` – the original CSV files from the enriched VOC dataset (not committed due to file size restrictions).
- `data_clean/` – processed and filtered data (1700–1780), saved as zip archives for easier sharing.
- `tables/` – CSV summary tables generated during analysis (e.g., outcome rates, death vs recruitment by rank).
- `figures/` – exported visualizations used in the presentation and report.
- `notebooks/` – Python scripts for each stage of the workflow (`01_cleaning.py`, `02_descriptives.py`, `03_models.py`).
- `docs/` – project documentation files, including the codebook, method notes, model results, and figure captions.

Each script and output is labeled systematically, and the code has been written in modular form so that each step can be rerun independently. This makes it possible for future researchers or students to replicate or extend the study without altering the original structure.

Reproducibility

Anyone with Python (version 3.11 or higher) can reproduce the analysis by following these steps:

1. Clone or download the GitHub repository.
2. Place the raw dataset files inside the `data_raw/` folder.
3. Create and activate a virtual environment, then install dependencies listed in `requirements.txt`.
4. Run the scripts in the following order:
 - a. `01_cleaning.py` – prepares and filters the data;
 - b. `02_descriptives.py` – generates descriptive tables and figures;
 - c. `03_models.py` – performs machine learning analysis and produces model results.

All generated outputs will appear automatically in `data_clean/`, `tables/`, `figures/`, and `docs/`.

Long-term sustainability

The repository uses plain-text formats (CSV, TXT, MD) and open-source libraries to ensure long-term accessibility.

The structure allows for future reuse in several ways:

- Educational reuse: Future Digital Humanities students can rerun the scripts as a guided example of quantitative historical analysis.
- Research extension: New data, such as Asian or local recruits or voyage information, can be merged into the existing pipeline without major modification.
- Comparative studies: The same workflow can be applied to datasets from other trading companies, such as the English East India Company, enabling cross-company comparison of workforce structures.

All outputs are stored in widely readable formats and clearly referenced in the codebook.

The repository is currently private, as it is a part of a graded university project, but it can be made available upon request to the course instructor or future students for educational reuse.

Reflection

While an earlier establishment of the definitive Research Question (RQ) might have seemed more efficient—potentially preventing the initial literature review conducted by Emily and Dunya from being seen as "wasted"—we ultimately contend that every step in this project's process was meaningful and contributed to our final understanding.

The necessity of meeting the project proposal deadline was invaluable, as it successfully forced us to pivot and commit to a proper, feasible research direction. To enhance efficiency in future collaborative efforts, a key improvement would be ensuring all team members possess a basic foundational knowledge of coding, as time was spent by Feruza explaining technical concepts to Emily and Dunya to establish shared understanding.

Looking ahead, this project opens up several compelling directions for future research: enriching the dataset by integrating it with external data (such as financial or shipping logs);

dedicating more time to find and model complex statistical patterns; conducting a robust comparative analysis with the British East India Company (EIC); formally establishing interdisciplinary collaboration with historians to strengthen contextual grounding; and expanding the literature review to engage more deeply with historical scholarship.

Finally, a significant personal lesson learned was the ability to effectively work with individuals from disparate academic backgrounds. Though our initial ways of thinking—one prioritizing computational accuracy, the others valuing social context—were very different, this dynamic was successfully managed through open discussion and a shared commitment to listening. We are fortunate that the kindness and willingness of all teammates to engage made this challenging collaborative process a successful, unifying experience.